Out look	Temperature	Hot	Windy	Rain
1- Sunny	Hot	High	Weak	No
2- Sunny	Mild	High	Strong	No
3- Cloudy	Hot	High	Weak	Yes
4- Rainy	Mild	High	Weak	Yes
5- Rainy	Cool	Normal	Weak	Yes
6- Rainy	Cool	Normal	Strong	No
7- Cloudy	Mild	Normal	Strong	Yes
8- Sunny	Cool	Normal	Weak	No

Total Entropy:

$$= -\left(\overbrace{\frac{4}{8}\log_2\frac{4}{8}}^{\text{Yes}} - \overbrace{\frac{4}{8}\log_2\frac{4}{8}}^{\text{No}}\right)$$

$$= -\left(0.5\log_2 0.5 - 0.5\log_2 0.5\right)$$

$$= -\left(0.5\left(\frac{\log 0.5}{\log 2}\right) - 0.5\left(\frac{\log 0.5}{\log 2}\right)\right)$$

$$= -\left(0.5\left(\frac{-0.301}{0.301}\right) - 0.5\left(\frac{-0.301}{0.301}\right)\right)$$

$$= -\left(0.5(-1) - 0.5(-1)\right)$$

$$= -(-0.5 - (-0.5))$$

$$= -(-0.5 + 0.5)$$

$$= 0 \quad \text{(Total Entropy)}$$

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Entropy of outlook:

Sunny:

	Yes	NO
(3) Sunny	0	3

$$= \frac{0}{3}\log_2\frac{0}{3} + \frac{3}{3}\log_2\frac{3}{3}$$

$$= 0 + 1\log_2 1$$

$$= 0 + 1\left(\frac{\log 1}{\log 2}\right)$$

$$= 0$$

Cloudy:

	Yes	NO
(2) cloudy	2	0

$$= \frac{2}{2}\log_2\frac{2}{2} + \frac{0}{2}\log_2\frac{0}{2}$$

$$= 1\log_2 1 + 0$$

$$= 0$$

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Rainy:	Yes	NO
(3) Rainy	2	1

$$= \frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}$$

$$= 0.666\left(\frac{\log(0.666)}{\log 2}\right) + 0.333\left(\frac{\log 0.333}{\log 2}\right)$$

$$= 0.666\left(\frac{-0.176}{0.301}\right) + 0.333\left(\frac{-0.477}{0.301}\right)$$

$$= 0.666\,(-0.584) + (0.333\,(-1.584))$$

$$= -(0.388) - 0.527)$$

$$= -0.915$$

Weighted Entropy of outlook:

$$= \frac{3}{8}(0) + \frac{2}{8}(0) + \frac{3}{8}(-0.915)$$

$$= -0.343$$

$$= 0 - (-0.343)$$

$$= 0.343$$

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Temperature:

	Yes	NO
(2) Hot	1	1
(3) Mild	2	1
(3) Cool	1	2

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